

Resolve lambda init-capture pack grammar (CWG2378)

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§ 1 Introduction

When [\[P0780R2\]](#) was adopted in Jacksonville in 2018, the paper contained a grammar rule that was inconsistent with other uses of packs. As pointed out by Richard Smith a few hours after plenary [\[Smith.Core\]](#), the grammar requires a pack of references to be captured like:

```
[...&x=init]
```

rather than how we write packs of references everywhere else.

```
[&...x=init]
```

This is now [\[CWG2378\]](#).

As of this writing, clang implements the Core Issue direction (with the `&` preceeding the `...`) and gcc implements the standard wording. I opened a gcc bug for this [\[gcc.91847\]](#), but we should really fix the wording asap.

This paper exists to do that.

§ 2 Wording

Change the grammar in 7.5.5.2 [\[expr.prim.lambda.capture\]](#):

```
capture:
- simple-capture ...opt
- ...opt init-capture
+ simple-capture
+ init-capture

simple-capture:
- identifier
- & identifier
+ identifier ...opt
+ & identifier ...opt
```

```

    this
    * this

    init-capture:
-   identifier initializer
-   & identifier initializer
+   ...opt identifier initializer
+   & ...opt identifier initializer

```

Change 7.5.5.2 [expr.prim.lambda.capture]/2:

If a *lambda-capture* includes a *capture-default* that is `=`, each *simple-capture* of that *lambda-capture* shall be of the form “& `...opt` *identifier*”, “`this`”, or “`* this`”.

Change 7.5.5.2 [expr.prim.lambda.capture]/6:

An *init-capture* `without ellipsis` behaves as if it declares and explicitly captures a variable of the form “`auto init-capture ;`” whose declarative region is the *lambda-expression*’s compound-statement, except that:

Change 7.5.5.2 [expr.prim.lambda.capture]/17:

- ¹⁷ A *simple-capture* `followed by containing` an ellipsis is a pack expansion ([temp.variadic]). An *init-capture* `preceded by containing` an ellipsis is a pack expansion that introduces an *init-capture* pack ([temp.variadic]) whose declarative region is the *lambda-expression*’s compound-statement.

Change 13.7.3 [temp.variadic]/5.10:

- ⁵ [...] Pack expansions can occur in the following contexts:
- (5.10) — In a *capture-list* ([expr.prim.lambda]); the pattern is ~~a-capture~~ `the capture without the ellipses`.

§ 3 References

[CWG2378] Barry Revzin. 2018. Inconsistent grammar for reference init-capture of pack.
<https://wg21.link/cwg2378>

[gcc.91847] Barry Revzin. 2019. init-capture pack of references requires ... on wrong side.
https://gcc.gnu.org/bugzilla/show_bug.cgi?id=91847

[P0780R2] Barry Revzin. 2018. Allow pack expansion in lambda init-capture.
<https://wg21.link/p0780r2>

[Smith.Core] Richard Smith. 2018. p0780r2 has wrong grammar for reference init-capture packs.
<https://lists.isocpp.org/core/2018/03/4095.php>